

Aashish Singhal

Senior Full-Stack Engineer

aashish451singhal@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [+91 9672080577](#)

SUMMARY

Senior Full-Stack Engineer with 5+ years shipping production web applications in React, TypeScript, and Go. Specialized in data-dense product surfaces — analytics workspaces, review workflows, and AI-powered dashboards — backed by distributed services and cloud infrastructure. Takes features from zero to production.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Python, SQL (PostgreSQL), MongoDB, ClickHouse, Redis

Frameworks & Tools: React, Next.js, Node.js, TailwindCSS, gRPC, Protobuf, Temporal, WebSockets, REST APIs, GraphQL, Grafana

AI & LLM: LLM APIs (Claude, GPT, Gemini), LLM Evaluations, Prompt Engineering, Langfuse, OpenAI Whisper, Deepgram

Infrastructure: Kubernetes (GKE), Docker, Google Cloud Platform, Amazon Web Services, Cloudflare Workers, CI/CD, GitHub Actions

EXPERIENCE

Senior Software Engineer | 100ms — AI call-quality platform Mar 2026 – Present *Bangalore, India*

- Product Engineering:** Shipped an AI call-quality review platform from zero in **Go** and **React/TypeScript** — role-gated evaluation list, transcript viewer with search and timestamp navigation, scorecard UI, and a human-in-the-loop accept/reject workflow that recomputes scores from reviewer answers.
- Analytics & Performance:** Built an org/agent analytics workspace with trend bucketing, RBAC, and CSV/PDF export; eliminated production API timeouts by re-scoping aggregations to page date bounds and fixing index-hostile sorts.
- Large-Scale Data Pipelines:** Designed and operated a production backfill (stitch → transcribe → LLM evaluation → delivery) processing **106K+** historical call recordings on **Google Kubernetes Engine**, delivering **92K+** AI evaluations at **98%** completion with every failed row reconciled.
- LLM Evaluation:** Replaced rule-based compliance and escalation detection with **LLM transcript analysis**; hardened prompts and output schemas over 10+ iterations to guarantee timestamp-anchored, in-bounds evidence. Built a **Langfuse** dataset bridge closing the production-to-eval-set loop, and ran 20-iteration consistency experiments to measure prompt stability before shipping.

Software Engineer | Empirical.run — autonomous software-testing agents Apr 2025 – Jan 2026 *Gurugram, India*

- Product Development:** Delivered Linear integration (OAuth, webhooks), GitHub PR management, authentication systems (Magic Links, TOTP), and a mobile-responsive React/Next.js dashboard with automated test triage.
- AI/LLM Infrastructure:** Engineered production multi-model LLM infrastructure (**Claude**, **GPT-5**, **Gemini**) with canonical message transformation, Zod v4 schema validation, and Rollbar error tracking across **200+ PRs**.
- Distributed Systems:** Architected a **Cloudflare Durable Objects** agent worker with **WebSocket** streaming, message queuing, and SQL state management; built the production observability stack (Langfuse tracing, PostHog, Rollbar) and CI/CD with ECS health checks and preview environments.

Member of Technical Staff 2 | Nutanix — hybrid cloud infrastructure Jan 2022 – Apr 2025 *Bangalore, India*

- Integrated OS version deployment behind feature flags for phased rollout to the Sales portal, reducing deployment failures by **40%**; designed the feature-flag portal that improved release efficiency by **25%**.
- Migrated the application frontend to a new in-house UI library, cutting UI-related bug reports by **30%**; built User Permission Manager and Bulk Edit features with **React** and **GraphQL**, increasing workflow automation by **20%**.

Full-Stack Development Intern | Celebal Technologies — data & enterprise cloud Jan 2021 – Dec 2021 *Jaipur, India*

- Developed order-management and HR portals with a **50%** performance improvement; mentored interns on React, Redux, and TypeScript.

PROJECTS

Videostil — Open Source | *TypeScript, Node.js, FFmpeg, LLM APIs* Oct 2025

- Open-sourced a production video pipeline that extracts and deduplicates frames for **LLM context windows**, supporting Claude, GPT, and Gemini APIs across **13 releases** on **npm**.
- Engineered three deduplication algorithms (greedy, dynamic programming, sliding window) with FFmpeg integration, frame normalization, and configurable FPS extraction.

AI Incident Triage | *NestJS, React, DynamoDB, OpenAI, AWS CDK* Jan 2026

- Built a three-tier triage system: a **NestJS** API that enriches incoming incidents with **OpenAI**-generated summaries and suggested actions, persists to **DynamoDB**, and serves a React/TanStack review UI.
- Provisioned infrastructure as code with **AWS CDK**, emulated AWS locally with LocalStack, and automated build and deploy across three **GitHub Actions** workflows.

totp-web — npm Library | *TypeScript, Web Crypto API* Apr 2025

- Published a zero-dependency **TOTP** two-factor authentication generator and verifier built on the **Web Crypto API**, shipping a typed API, CLI, test suite, and automated semantic releases via changesets.

EDUCATION

Poornima College of Engineering | B.Tech, Computer Science

Jaipur, India 2017 – 2021